DSATM Intelligent Systems Hackathon

EchoRoute

Intelligent System for Real-Time Smart Commuting Decisions

Team ASTRA 🔱

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DOMAIN

> Intelligence systems for real world decision making.

TEAM MEMBERS

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EchoRoute Project Logo

The Problem Statement

Unpredictable Commutes

Daily commuters face highly unpredictable traffic patterns, resulting in massive time loss due to poor, uninformed route choices made before or during the journey.

Limited Existing Solutions

Current systems primarily focus only on the shortest or fastest route. They completely fail to consider multiple real-world factors concurrently to make a truly intelligent decision.

Problem Impact

Wasted Travel Time

Commuters lose hundreds of hours annually stuck in gridlocks.

Increased Stress

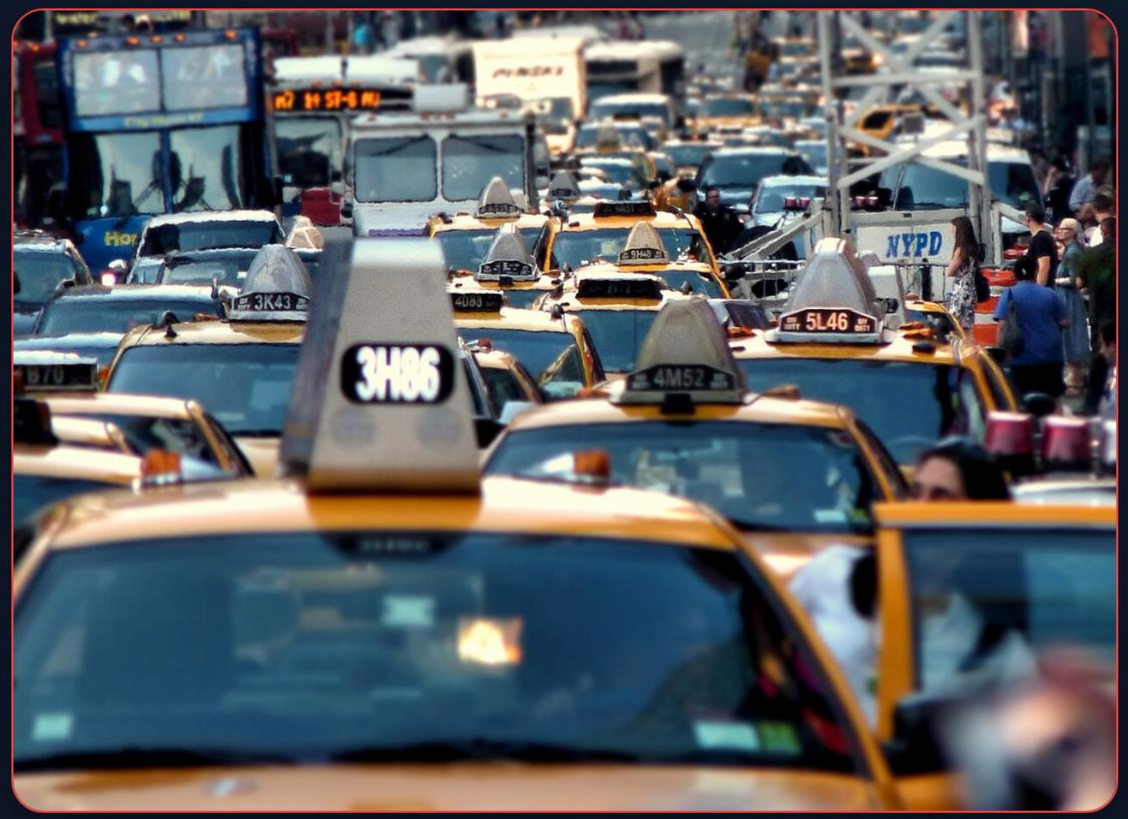
Unpredictable delays lead to high frustration and reduced productivity.

Higher Fuel Consumption

Idling in traffic significantly burns fuel and increases costs.

Poor Urban Experience

Decreases the overall quality of living and navigating in modern cities.



Our Solution

EchoRoute is an intelligent system designed to revolutionize urban travel.

It suggests optimal commuting routes in **real time** by deeply considering dynamic traffic conditions and live environmental data.

By shifting from simple mapping to **smarter decision-making**, EchoRoute ensures the most efficient, low-stress travel experience possible.

System Architecture

User Input

User inputs their desired destination into the interface.

Data Collection

Live gathering of current traffic, route, and environmental info.

Decision Engine

Intelligent algorithm weighs parameters and analyzes routes.

Best Route Output

The smartest, highly-optimized path is delivered to the user.

Key Features

Real-Time Route Optimization

Instantly adapts to changing road conditions as they happen.

Traffic-Aware Decision Making

Understands density, blockages, and delays beyond simple distance.

Smart Route Comparison

Weighs multiple paths simultaneously to find the absolute best option.

Simple User Interface

Clean, intuitive design ensures any commuter can use it effortlessly.

Fast Response System

Low-latency architecture guarantees split-second routing updates.

How It Works

1

Destination Entry

The user simply enters their final destination into the EchoRoute platform.

2

Data Gathering

The system immediately pulls real-time route geometries and live traffic density data.

3

Route Analysis

Multiple potential routes are concurrently analyzed by our intelligent weighting algorithm.

4

Optimization

The best, most optimized route is selected and actively guided for the user.

Technology Stack

COMPONENT	TECHNOLOGIES USED	PURPOSE
Frontend	React / Flutter	Ensures a simple, responsive, and intuitive User Interface.
Backend	Node.js / Python	Handles rapid data processing and system architecture.
APIs	Google Maps / Traffic APIs	Provides real-time geospatial, mapping, and live traffic data.
Logic	Decision-making Algorithm	Calculates dynamic weights for intelligent routing decisions.

Core Innovation

Beyond the Shortest Path

EchoRoute looks past simple distance metrics. It seeks the **smartest path** by simulating intelligent, human-like commuting choices under complex conditions.

Scalable & Real-Time

Built completely on real-time condition evaluation, the underlying architecture is highly scalable and ready for integration into broader Smart City ecosystems.

Live Demo

Watch EchoRoute's intelligent routing in action.

Saves Time

Improves Efficiency

Reduces Stress

Thank You!

Team ASTRA

Image Sources

Thumbnail for www.freepik.com

https://img.freepik.com/premium-psd/green-eco-location-map-pin-icon_1050938-29187.jpg?semt=ais_hybrid&w=740&q=80

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